1 Gain law

(bars denote normalized quantities: lengths by R; $a_o := \Delta t/(\gamma_N R)$ [1/pN] and $a_{\text{nom}} := \Delta t/\gamma_N = a_o R$ [μm/pN]; $\bar{a} := a/a_{\text{nom}}$; forces by a_o: $\bar{f} := a_o f$, so that $\bar{a}\,\bar{f} = \Delta\bar{w}$; $0 \le \bar{a} \le 1$, far field 1, contact 0; hats sit on the normalized quantities $\hat{b}$, $\delta\hat{\bar{w}}$; $e_\theta = \theta - \hat{\theta}$; $\bar{w}_s$ known)

$$
\begin{aligned}
\bar{a}(\bar{w}, b) &= 1 - \frac{1}{b\,(\bar{w} - \bar{w}_s)} \\
\bar{a}'(\bar{a}, b) &:= \frac{\partial \bar{a}}{\partial \bar{w}} = b\big(1 - \bar{a}\big)^2 \\
\Big(\frac{\partial \bar{a}}{\partial b}\Big)_{\bar{w},\,\bar{w}_s} &= \frac{1 - \bar{a}}{b} \\
a_w = a_o\,\bar{a}_w, &\qquad a_o = \frac{\Delta t}{\gamma_N R}
\end{aligned}
$$

2 True dynamics

$\big(\delta\bar{w}_j[k] := \delta\bar{w}[k - d + j - 1],\ j = 1, \ldots, d+1,\ d = 2\big)$

$$
\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k] \\
\bar{w}_d[k+1] &= \bar{w}_d[k] + \Delta\bar{w}_d[k] \\
\bar{a}_w[k] &= 1 - \frac{1}{b\,\big(\bar{w}[k] - \bar{w}_s\big)} \\
\delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k] \\
\delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\Delta\bar{w}[k] &= \Delta\bar{w}_d[k] + (1 - \lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
b[k+1] &= b[k]
\end{aligned}
$$

$$
F_{dw}[k] = \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^{d} \bar{f}_{dw}[k - i]
$$

$$
\varepsilon_w[k] = \bar{a}_w[k]\,\bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^{d} \bar{a}_w[k - i]\,\bar{f}_T[k - i] - (1 - \lambda_c)\,n_w[k]
$$

3 Estimator

$$
\mathbf{x} = \big[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ b\,\big]^\top \qquad (d = 2)
$$

$$
\begin{aligned}
\hat{\bar{w}}[k] &= \bar{w}_d[k] - \delta\hat{\bar{w}}_3[k] \\
\hat{\bar{a}}_w[k] &= 1 - \frac{1}{\hat{b}[k]\,\big(\hat{\bar{w}}[k] - \bar{w}_s\big)} \\
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k] \\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k] \\
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k] \\
\hat{b}[k+1] &= \hat{b}[k]
\end{aligned}
$$

4 Error dynamics

$\big(\mathbf{e}[k] = [\, e_{\delta\bar{w}_1},\ e_{\delta\bar{w}_2},\ e_{\delta\bar{w}_3},\ e_b\,]^\top$, coefficients at the estimate$\big)$

$$
\begin{aligned}
\bar{w}[k] - \hat{\bar{w}}[k] &= -\,e_{\delta\bar{w}_3}[k] \\
e_{\bar{a}_w}[k] &= \bar{a}\big(\bar{w}[k], b\big) - \bar{a}\big(\hat{\bar{w}}[k], \hat{b}[k]\big) = -\,\bar{a}'[k]\,e_{\delta\bar{w}_3}[k] + \frac{1 - \hat{\bar{a}}_w[k]}{\hat{b}[k]}\,e_b[k]
\end{aligned}
$$

$$
\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \Big(\lambda_c + F_{dw}[k]\,\bar{a}'[k]\Big)e_{\delta\bar{w}_3}[k] - F_{dw}[k]\,\frac{1 - \hat{\bar{a}}_w[k]}{\hat{b}[k]}\,e_b[k] - \varepsilon_w[k] \\
e_b[k+1] &= e_b[k]
\end{aligned}
$$

$$
F_e[k] = \begin{bmatrix}
0 & 1 & 0 & 0 \\
0 & 0 & 1 & 0 \\
0 & 0 & \lambda_c + F_{dw}[k]\,\hat{b}[k]\big(1 - \hat{\bar{a}}_w[k]\big)^2 & -F_{dw}[k]\,\dfrac{1 - \hat{\bar{a}}_w[k]}{\hat{b}[k]} \\
0 & 0 & 0 & 1
\end{bmatrix}
$$

5 Measurement

$\big(\nabla_d\bar{w}_d[k] := \bar{w}_d[k] - \bar{w}_d[k-d]\big)$

$$
\begin{aligned}
y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_w[k-d] + n_a[k], \qquad \bar{a}_w[k-d] = 1 - \frac{1}{b\left(\bar{w}[k-d] - \bar{w}_s\right)} \\
\hat{y}_1[k] &= \delta\hat{\bar{w}}_1[k] \\
\hat{\bar{w}}_d[k] &= \hat{\bar{w}}[k] - \nabla_d\bar{w}_d[k] \\
\hat{\bar{a}}_d[k] &= 1 - \frac{1}{\hat{b}[k]\left(\hat{\bar{w}}_d[k] - \bar{w}_s\right)} \\
\hat{y}_2[k] &= \hat{\bar{a}}_d[k] \\
H_{23}[k] &= \frac{\partial\hat{y}_2[k]}{\partial\delta\hat{\bar{w}}_3[k]} = -\hat{b}[k]\big(1 - \hat{\bar{a}}_d[k]\big)^2 \\
H_{24}[k] &= \frac{\partial\hat{y}_2[k]}{\partial\hat{b}[k]} = \frac{1 - \hat{\bar{a}}_d[k]}{\hat{b}[k]} \\
H[k] &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & H_{23}[k] & H_{24}[k] \end{bmatrix} \\
\mathbf{e}_y[k] &= H[k]\,\mathbf{e}^-[k] + \begin{bmatrix} n_w[k] \\ n_a[k] \end{bmatrix}
\end{aligned}
$$

6 Noise

$$
\begin{aligned}
Q[k] &= Q_{33}[k]\,\mathbf{g}\,\mathbf{g}^\top, \qquad \mathbf{g} = \begin{bmatrix} 0,\ 0,\ -1,\ 0 \end{bmatrix}^\top, \qquad Q_{44} = 0 \\
Q_{33}[k] &= \kappa_T\Big(\hat{\bar{a}}_w[k] + (1-\lambda_c)^2\big(\hat{\bar{a}}_w[k-1] + \hat{\bar{a}}_w[k-2]\big)\Big) + (1-\lambda_c)^2\sigma_{n_w}^2, \qquad \kappa_T = \frac{4k_BT\,a_o}{R} \\
R[k] &= \mathrm{diag}\big(R_1,\ R_2[k]\big), \qquad R_1 = \sigma_{n_w}^2 = \frac{\sigma_n^2}{R^2}, \qquad R_2[k] = 2a_{\mathrm{cov}}^2\,IF_{\mathrm{var}}\big(\hat{\bar{a}}_w[k]\big)\big(\hat{\bar{a}}_w[k] + \xi\big)^2
\end{aligned}
$$

7 Observability

($\Phi(k,k_0) := F_e[k-1]\,F_e[k-2]\cdots F_e[k_0]$, $\Phi(k_0,k_0)=I$; N window length; $O_w := R_w^{-1/2}\,[\,H[k_0]\Phi(k_0,k_0);\;\ldots;\;H[k_0+N]\Phi(k_0+N,k_0)\,]$; $O_w\,\mathbf{v}_i = \sigma_i\,\mathbf{u}_i$, $\sigma_1 \geq \cdots \geq \sigma_4$)

$$F_e[k]_{:,4} = \Big[\,0,\;0,\;-F_{dw}[k]\,\frac{1-\hat{\bar{a}}_w[k]}{\hat{b}[k]},\;1\,\Big]^\top$$
$$H[k]_{:,4} = \Big[\,0,\;\frac{1-\hat{\bar{a}}_d[k]}{\hat{b}[k]}\,\Big]^\top$$

$$\Delta\bar{w}_d[k]=0\;\;\forall k \quad\Longrightarrow\quad F_{dw}[k]\to 0, \qquad \nabla_d\bar{w}_d[k]=0$$
$$\Longrightarrow\quad F_e[k]_{:,4} = \big[\,0,\;0,\;0,\;1\,\big]^\top, \qquad H[k]_{:,4} = \Big[\,0,\;\frac{1-\hat{\bar{a}}_w[k]}{\hat{b}[k]}\,\Big]^\top \neq \mathbf{0}$$

$\operatorname{rank} O_w = 4$

$$\sigma_b := \sqrt{\big[\,(O_w^\top O_w)^{-1}\,\big]_{44}}$$

Seed and prior

$$\hat{b}[0] = \frac{1/\big(1-\bar{a}_w[0]\big)-1}{\bar{w}[0]-\bar{w}_c}, \qquad \bar{w}_s = \bar{w}_c - \frac{1}{\hat{b}[0]}$$
$$P[0] = \operatorname{diag}\big(P_{11}[0],\;P_{22}[0],\;P_{33}[0],\;P_{44}[0]\big), \qquad \sqrt{P_{44}[0]} = \sigma_{b,0}$$
$$\sqrt{\operatorname{Var}\big(\hat{\bar{a}}_w[0]\big)} = \frac{1-\hat{\bar{a}}_w[0]}{\hat{b}[0]}\,\sigma_{b,0}$$